Mechanical Engineering Experiments — Report 3

機械工程實驗(一) 第三次報告

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Q1:

(a) .

Original length, l = 155.05mm

Width, w = 19.95mm

Thickness, t = 0.45mm

Weight, m = 9.68g

Clipped length, L = 137mm

$$I = \frac{w \times t^3}{12} = \frac{19.95 \times 0.45^3}{12} \approx 0.1515mm^4 = 0.1515 \times 10^{-12}m^4$$

$$A = w \times t = 19.95 \times 0.45 \approx 8.978mm^2 = 8.978 \times 10^{-6}m^2$$

$$\rho = \frac{m}{l \times A} = \frac{0.00968}{155.05 \times 8.978} \approx 6.954 \times 10^{-6}kg/mm^3 = 6.954 \times 10^{3}kg/m^3$$

$$\omega_1 = \frac{(\beta_1 L)^2}{L^2}\sqrt{\frac{EI}{\rho A}} = \frac{1.875^2}{(137 \times 10^{-3})^2}\sqrt{\frac{208000 \times 10^6 \times 0.1515 \times 10^{-12}}{6.954 \times 10^3 \times 8.978 \times 10^{-6}}} \approx 133.073rad/s$$

$$\omega_2 = \frac{(\beta_2 L)^2}{L^2}\sqrt{\frac{EI}{\rho A}} = \frac{4.694^2}{(137 \times 10^{-3})^2}\sqrt{\frac{208000 \times 10^6 \times 0.1515 \times 10^{-12}}{6.954 \times 10^3 \times 8.978 \times 10^{-6}}} \approx 834.017rad/s$$

$$\omega_3 = \frac{(\beta_3 L)^2}{L^2}\sqrt{\frac{EI}{\rho A}} = \frac{7.855^2}{(137 \times 10^{-3})^2}\sqrt{\frac{208000 \times 10^6 \times 0.1515 \times 10^{-12}}{6.954 \times 10^3 \times 8.978 \times 10^{-6}}} \approx 2335.507rad/s$$

$$\omega_4 = \frac{(\beta_4 L)^2}{L^2}\sqrt{\frac{EI}{\rho A}} = \frac{11^2}{(137 \times 10^{-3})^2}\sqrt{\frac{208000 \times 10^6 \times 0.1515 \times 10^{-12}}{6.954 \times 10^3 \times 8.978 \times 10^{-6}}} \approx 4580.092rad/s$$

(b) .

Table1. Resonant frequencies of the buzzer

	n^{th}	$n+1^{th}$	$n+2^{th}$
Resonant frequency (Hz)	3290	3805	11520
Peak-to-peak voltage (V)	7.32	6.59	0.556

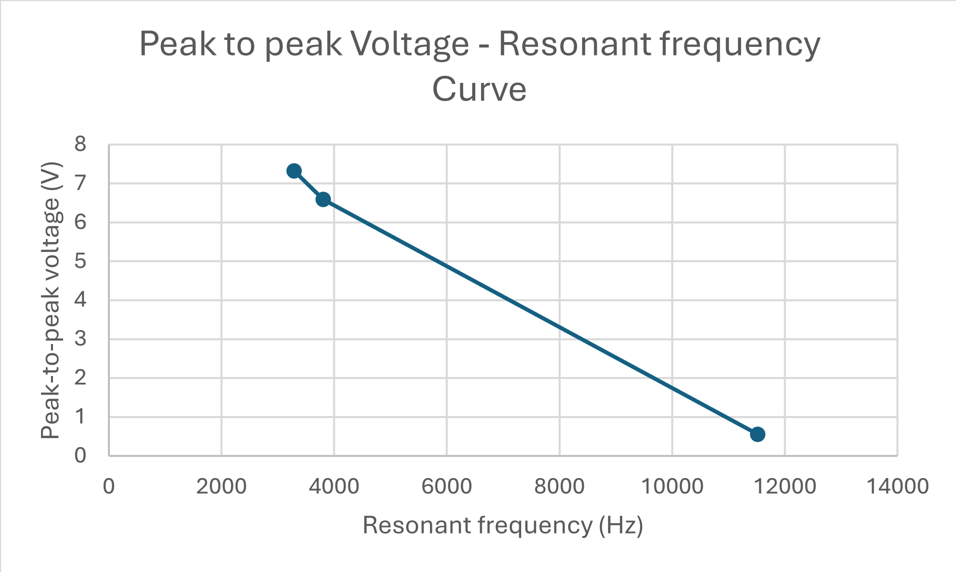

Figure1. Peak to peak Voltage - Resonant frequency Curve

The buzzer shows different resonant frequencies, but the peak-to-peak voltage decreases for higher resonant frequency. This is mainly because higher frequency has more complex mode shapes and higher effective stiffness, so the local strain at the sensor location is small, therefore the piezoelectric output is smaller.

(c) .

For experiment:

Table2. Resonant frequencies of the cantilever beam

	1^{st}	2^{nd}	3^{rd}
Resonant frequency (Hz)	114.5	319.1	622.4

For theory:

$$f_n = \frac{\omega_n}{2\pi}$$

$$f_1 = \frac{\omega_1}{2\pi} = \frac{133.073}{2\pi} \approx 21.179Hz$$

Because the experiment starts at frequency = 70Hz wave, we didn't measure the 1^{st} resonant frequency in theory. We can compare the f_2 with the 1^{st} data we collected and f_3 with the 2^{nd} data…

$$f_{theory,1} = f_2 = \frac{\omega_2}{2\pi} = \frac{834.017}{2\pi} \approx 132.738Hz$$

$$f_{theory,2} = f_3 = \frac{\omega_3}{2\pi} = \frac{2335.507}{2\pi} \approx 371.707Hz$$

$$f_{theory,3} = f_4 = \frac{\omega_4}{2\pi} = \frac{4580.092}{2\pi} \approx 728.944Hz$$

$$Measurement\ error_1 = \frac{f_{exp,1} - f_{theory,1}}{f_{theory,1}} = \frac{114.5 - 132.738}{132.738} \approx -13.740\%$$

$$Measurement\ error_2 = \frac{f_{exp,2} - f_{theory,2}}{f_{theory,2}} = \frac{319.1 - 371.707}{371.707} \approx -14.153\%$$

$$Measurement\ error_3 = \frac{f_{exp,3} - f_{theory,3}}{f_{theory,3}} = \frac{622.4 - 728.944}{728.944} \approx -14.616\%$$

(d) .

Assume the right weight (m_{real}) is x% more than measured weight ($m_{measured}$) so that $m_{real} = m_{measured}(1 + x)$.

Since $\rho = \frac{m}{l \times A}$, $\rho_{real} = \rho_{measured}(1 + x)$.

$$\omega_{real,2} = \frac{(\beta_2 L)^2}{L^2}\sqrt{\frac{EI}{\rho_{measured}(1+x)A}} = \frac{(\beta_2 L)^2}{L^2}\sqrt{\frac{EI}{\rho_{measured}A}}\frac{1}{\sqrt{1+x}} =$$

$$\frac{4.694^2}{(137 \times 10^{-3})^2}\sqrt{\frac{208000 \times 10^6 \times 0.1515 \times 10^{-12}}{6.954 \times 10^3 \times 8.978 \times 10^{-6}}}\frac{1}{\sqrt{1+x}} \approx \frac{834.017}{\sqrt{1+x}} rad/s$$

(e) .

$$Measurement\ error_1 \approx -13.740\%$$

$$f_{real,1} = \frac{\omega_{real,2}}{2\pi} \approx \frac{132.738}{\sqrt{1+x}} Hz$$

$$Measurement\ error_1 = \frac{f_{real,1} - f_{theory,1}}{f_{theory,1}} \times 100\% = \frac{\frac{132.738}{\sqrt{1+x}} - 132.738}{132.738} \times 100\% \approx$$

$$-13.740\%$$

$$|x| \approx 0.344 > 5\% \rightarrow not\ reasonable$$

Q2:

(a) .

Table3. Voltage response from different accelerometers

frequency (Hz)	Vpp,homemade (V)	Vpp,commercial (V)	Vpp,home/Vpp,com	10*Vpp,home/Vpp,com (mv/g)
204.1	1.44	0.01766	81.54020385	815.4020385
250.8	1.29	0.01673	77.10699342	771.0699342
300.7	1.44	0.01605	89.71962617	897.1962617
348	1.33	0.01456	91.34615385	913.4615385
401.7	1.44	0.01455	98.96907216	989.6907216
450.2	1.36	0.01476	92.14092141	921.4092141
500.2	1.41	0.01389	101.511879	1015.11879
550.7	1.42	0.01428	99.43977591	994.3977591
601	1.39	0.01378	100.8708273	1008.708273
649.5	1.36	0.01384	98.26589595	982.6589595
705.7	1.41	0.01438	98.05285118	980.5285118
749.8	1.41	0.01337	105.459985	1054.59985
799.1	1.58	0.01301	121.4450423	1214.450423
848.7	1.65	0.01433	115.1430565	1151.430565
903.3	1.6	0.01321	121.1203634	1211.203634
948.3	1.63	0.01362	119.6769457	1196.769457
1007	1.72	0.01363	126.192223	1261.92223
1047	1.79	0.01425	125.6140351	1256.140351
1101	1.89	0.02349	80.45977011	804.5977011
1149	2.03	0.01384	146.6763006	1466.763006
1209	2.31	0.01253	184.3575419	1843.575419
1250	2.46	0.01174	209.5400341	2095.400341
1306	2.68	0.01104	242.7536232	2427.536232
1350	3	0.01016	295.2755906	2952.755906
1400	3.55	0.00817	434.5165239	4345.165239
1409	3.72	0.00771	482.4902724	4824.902724
1420	3.79	0.00722	524.9307479	5249.307479
1430	4	0.006	666.6666667	6666.666667
1439	4.03	0.00593	679.5952782	6795.952782
1447	4.19	0.00529	792.0604915	7920.604915
1459	4.48	0.00468	957.2649573	9572.649573
1469	4.76	0.00454	1048.45815	10484.5815
1479	5.11	0.0049	1042.857143	10428.57143
1480	6.07	0.0096	632.2916667	6322.916667
1481	5.84	0.00872	669.7247706	6697.247706
1482	6	0.00986	608.5192698	6085.192698
1483	6.07	0.0096	632.2916667	6322.916667
1485	6.15	0.01021	602.3506366	6023.506366
1488	6.32	0.01332	474.4744745	4744.744745
1490	7.03	0.01589	442.4166142	4424.166142
1491	9.52	0.03527	269.9177771	2699.177771
1492	9.35	0.03448	271.1716937	2711.716937
1493	10.2	0.04247	240.1695314	2401.695314
1496	10.5	0.04562	230.1622096	2301.622096
1500	10.7	0.0475	225.2631579	2252.631579
1505	11.6	0.05698	203.5802036	2035.802036
1550	13.3	0.07966	166.9595782	1669.595782
1599	11.4	0.09751	116.911086	1169.11086
1650	8.39	0.09014	93.0774351	930.774351
1699	6.23	0.08347	74.63759435	746.3759435
1749	4.71	0.07982	59.00776748	590.0776748
1806	3.59	0.0797	45.04391468	450.4391468

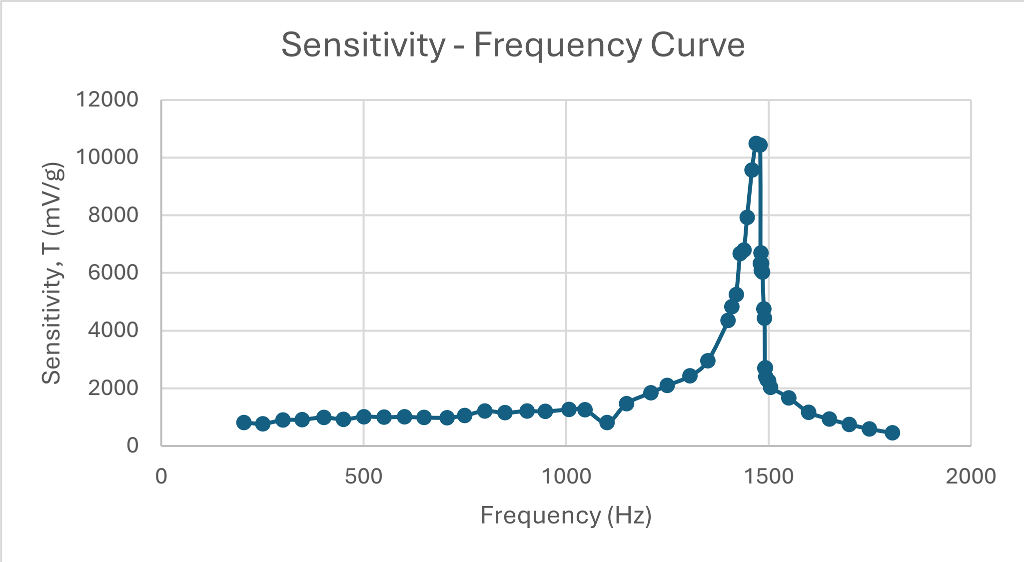

Figure2. Sensitivity – Frequency Curve

(b) .

Resonant frequency = the frequency at which sensitivity reaches its maximum

→ resonant frequency ≈ 1469Hz

For the shaking frequency << resonant frequency, the frequency should be relatively constant. We can take the data which frequency is less than 1047Hz into account and calculate its average:

$Constant\ sensitivity\ magnitude =$

$average\ of\ first\ 18\ data\ points\ (frequency < 1047Hz)\ in\ table3\ \approx$

$1035.342mV/g$

Q3.

In the formula $z_{particular}(t)\ /\ a\omega^2$, the numerator $z_{particular}(t)$ represent to the relative displacement z(t) = x(t) – y(t). When the base vibrates, the mass tends to stay “in place”, causing relative displacement z(t). Thus, the crystal is compressed or stretched depends on z(t) and since the piezoelectric material generates an electrical output voltage proportional to the mechanical compression ,z(t) directly corresponds to the crystal strain that generates the voltage output.

While the denominator $a\omega^2$ is directly related to the peak value of the applied acceleration. In the experiment, the shaker provides the base motion $a\sin(\omega t)$. Here, "a" represents the amplitude of the vibration, and "ω^2" represents to the frequency-dependent scaling of the acceleration. As the result, $a\omega^2$ normalizes the relative displacement $z(t)$ in terms of the applied acceleration magnitude.

In conclusion, the formula $z_{particular}(t) / a\omega^2$ represents the ratio between relative displacement (accelerometer's internal mass and vibrating mass) and base acceleration. And this ratio corresponds directly to the accelerometer's sensitivity, showing how the accelerometer converts base acceleration into an electrical signal.